

OCEAN LIFE PUBLIC AQUARIUM

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<http://classwork.engr.oregonstate.edu:1923/>

EXECUTIVE SUMMARY

Throughout the process of building our Ocean Life Public Aquarium database, we received feedback from TAs, peer reviewers, and the instructor. The biggest changes to our database design came from going through the normalization process. Considering feedback, we went from seven tables with eight entities to five tables and five entities. We had to focus our thinking and narrow the scope of the database to be more centered around a feeding log tracker for the busy aquarium.

The peer reviewers were a great way to spot and fix smaller errors. Smaller changes based on feedback included updating table and attribute names to follow naming conventions. We standardized VARCHAR lengths, updated the ER diagram to match the schema, fixed syntax errors, corrected typos, and added comments to the SQL code.

One peer review suggestion was to use CASCADE rules to properly manage deletions involving related records in our DDL file. Peer review on our user interface suggested that we use JOINS so that the UI would display names in the tables as opposed to just the raw foreign key IDs. Based on this peer review feedback, we implemented CASCADE rules in our SQL code and updated our SELECT queries to use JOINS.

Based on instructor feedback, we used ENUM data types to simplify attributes that had a fixed set of values. For example, in Employees, employment status could be one of three values, so we implemented ENUM.

Towards the final version of the project, we updated the DML and DDL SQL files and pulled the PL/SQL file out of them and added it to a separate PL file. We followed the class starter code and implanted a RESET button. Additionally, before the final submission we sanitized the password.

Microsoft Copilot AI was used to help with our SQL code and to review the final code against the grading rubric before submission. AI was useful for catching syntax and logic issues and helping us review code quickly. Any AI feedback was considered and reviewed by us.

OVERVIEW

Ocean Life is a public aquarium that is visited by approximately 3,000 people daily. The aquarium is home to more than 500 species including dolphins, tropical fish, octopuses, sharks, whales, penguins, seals, and rays [1][2]. The 15,000 marine animals at Ocean Life need to be fed every day and sometimes multiple times a day. The employees use various types of feed for the animals since they have different dietary needs. If these regular feeding operations are missed or performed inconsistently, the animal's health could quickly decline, or worse, the animals could lose their lives. Therefore, the aquarium's feeding operations need to be tracked thoroughly.

We propose a database management system to record and track these routine feeding events. By utilizing our database, the feed given to the animals and the employees responsible for certain feedings can be tracked. The database keeps track of the feeding events which were conducted by who, and when it was performed. Our database project will organize operations of the aquarium and help it become more efficient by keeping track of the feedings of the 15,000 animals.

OUTLINE

- **Employees**

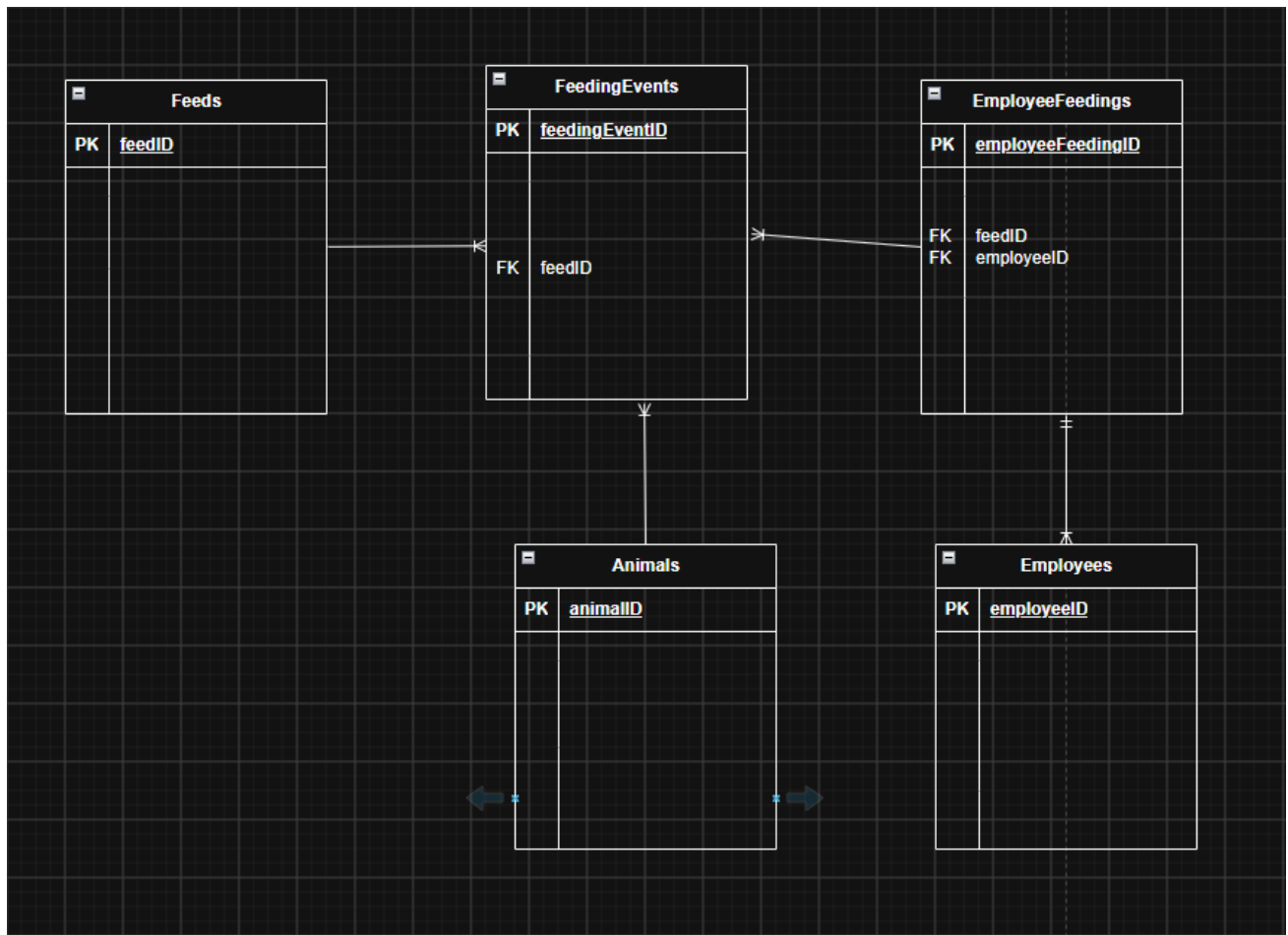
- ❖ employeeID: INT, auto_increment, unique, not NULL, PK
- firstName: VARCHAR(50), not NULL
- lastName: VARCHAR(50), not NULL
- employmentStatus: ENUM ('Part-time', 'Full-time', 'As-needed'), not NULL
- age: INT, not NULL
- Relationship: Many-to-many (M:N) with feedingEvents, each employee can do many feeding events, and each feeding event can involve multiple employees, Implemented with EmployeeFeedings intersection table.
- Overview: Holds the information of employee name, age, status as part time, full time or as needed.

- **FeedingEvents**

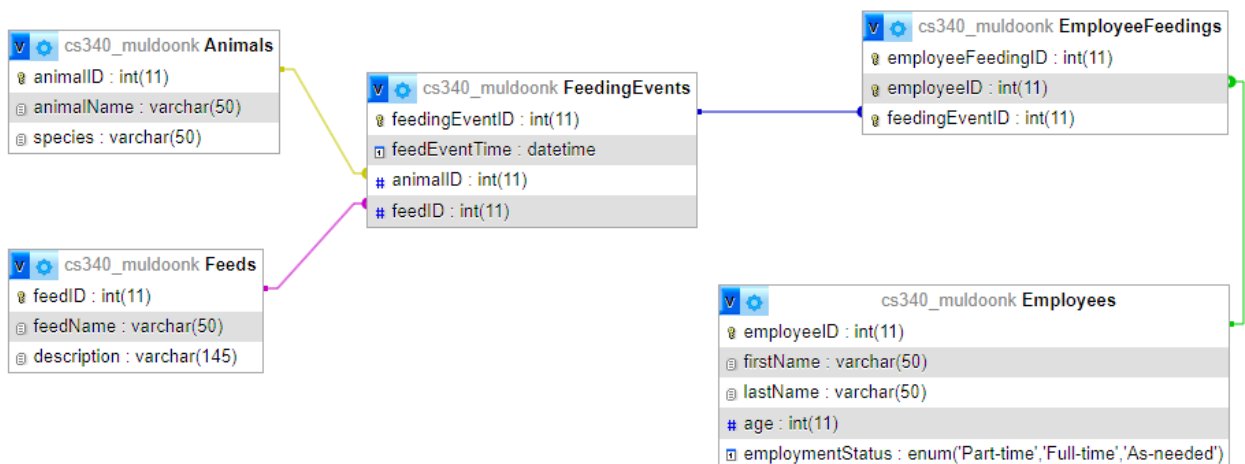
- ❖ feedingEventID: INT, auto_increment, unique, not NULL, PK
- feedEventTime: DATETIME, not NULL
- animalID: INT, not NULL, FK
- feedID: INT, not NULL, FK
- Relationship: Many-to-many (M:N) with Employees. Each employee can carry out multiple feeding events, and each animal feeding can have multiple employees.

- **Relationship:** Many-to-one (M:1) with Feeds. Many feeding events can use the same feed type while each feeding event uses one feed type.
 - **Relationship:** Many-to-one (M:1) with Animals. Many feeding events can be tracked for a single animal, and each feeding event is with one animal.
 - **Overview:** Each FeedingEvent is assigned a new id number and tracks what time and what food is being given to which animal by whom.
- **Animals**
 - ❖ **animalID:** INT, auto_increment, unique, not NULL, PK
 - **animalName:** VARCHAR(50), not NULL
 - **species:** VARCHAR(50), not NULL
 - **Relationship:** One-to-many (1:M) with FeedingEvents, an animal can be fed multiple times.
 - **Overview:** Gives each Animal its own ID so that animals of the same species can be identified by name.
- **Feeds**
 - ❖ **feedID:** INT, auto_increment, unique, not NULL, PK
 - **feedName:** VARCHAR(50), not NULL
 - **description:** VARCHAR(145), not NULL
 - **Relationship:** One-to-many (1:M) with FeedingEvents, each feeding event can use one feed type, but a feed type can be used in many different feedings.
 - **Overview:** Each feed type has its own id, name, and description for tracking and record keeping purposes.
- **EmployeeFeedings**
 - ❖ **employeeFeedingID:** INT, auto_increment, unique, not NULL, PK
 - **employeeID:** INT, not NULL, FK
 - **feedingEventID:** INT, not NULL, FK
 - **Relationship:** One-to-many (1:M) with Employees, each employee can appear in many EmployeeFeedings while each employee feeding event has one employee.
 - **Relationship:** One-to-many (1:M) with FeedingEvents, each feeding event can be on many EmployeeFeedings records while each EmployeeFeedings event is one feeding event.
 - **Overview:** Keeps track of the employees participating in feeding the animals by connecting employees with FeedingEvents.

ER DIAGRAM



SCHEMA DIAGRAM



EXAMPLE DATA

Animals Table — Sample Data

animalID	animalName	species
1	Gismo	Bottlenose Dolphin
2	Clover	Leopard Seal
3	Milo	Green Lanternshark
4	Kiki	East Pacific Red Octopus
5	Daisy	Green Sea Turtle

Employees Table - Sample Data

employeeID	firstName	lastName	age	employmentStatus
1	Steve	Irwin	45	Full-time
2	Robert	Jackson	22	Part-time
3	Elizabeth	Raynes	35	Full-time
4	Mary	Jones	20	Part-time

Feeds Table - Sample Data

feedID	feedName	description
1	Brine Shrimp Blend	A protein rich feed for carnivore animals.
2	Frozen Fish and Supplement Mix	A protein, vitamin and mineral rich feed for omnivore animals.
3	Plant Flake	A fiber rich feed for herbivore animals.

FeedingEvents Table - Sample Data

feedingEventID	feedEventTime	animalID	feedID
1	2026-07-21 09:20:00	1	2
2	2026-07-21 10:15:00	2	1
3	2026-07-21 10:40:00	3	1
4	2026-07-21 11:00:00	4	2
5	2026-07-21 11:00:00	5	3

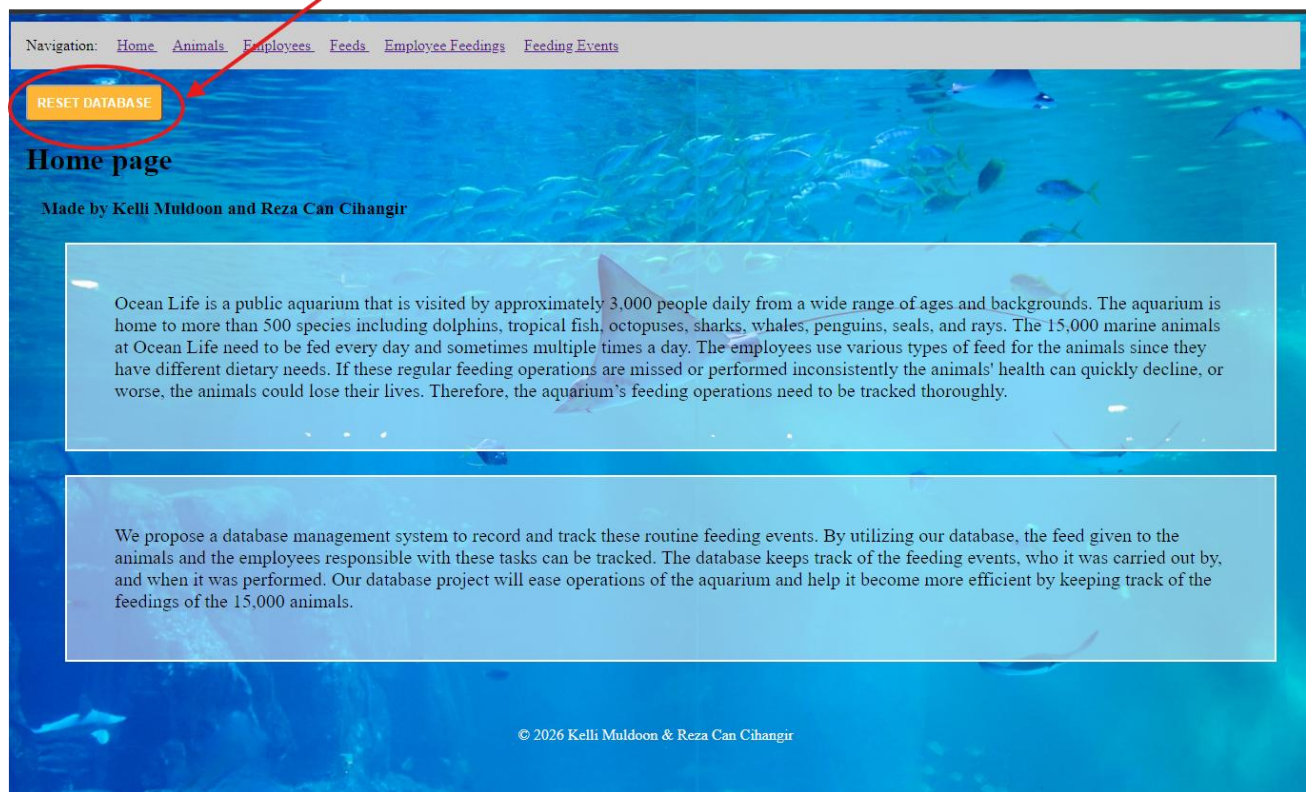
EmployeeFeedings Table - Sample Data

employeeFeedingID	employeeID	feedingEventID
1	1	1
2	2	1
3	1	2
4	3	3
5	4	3
6	4	4
7	2	5

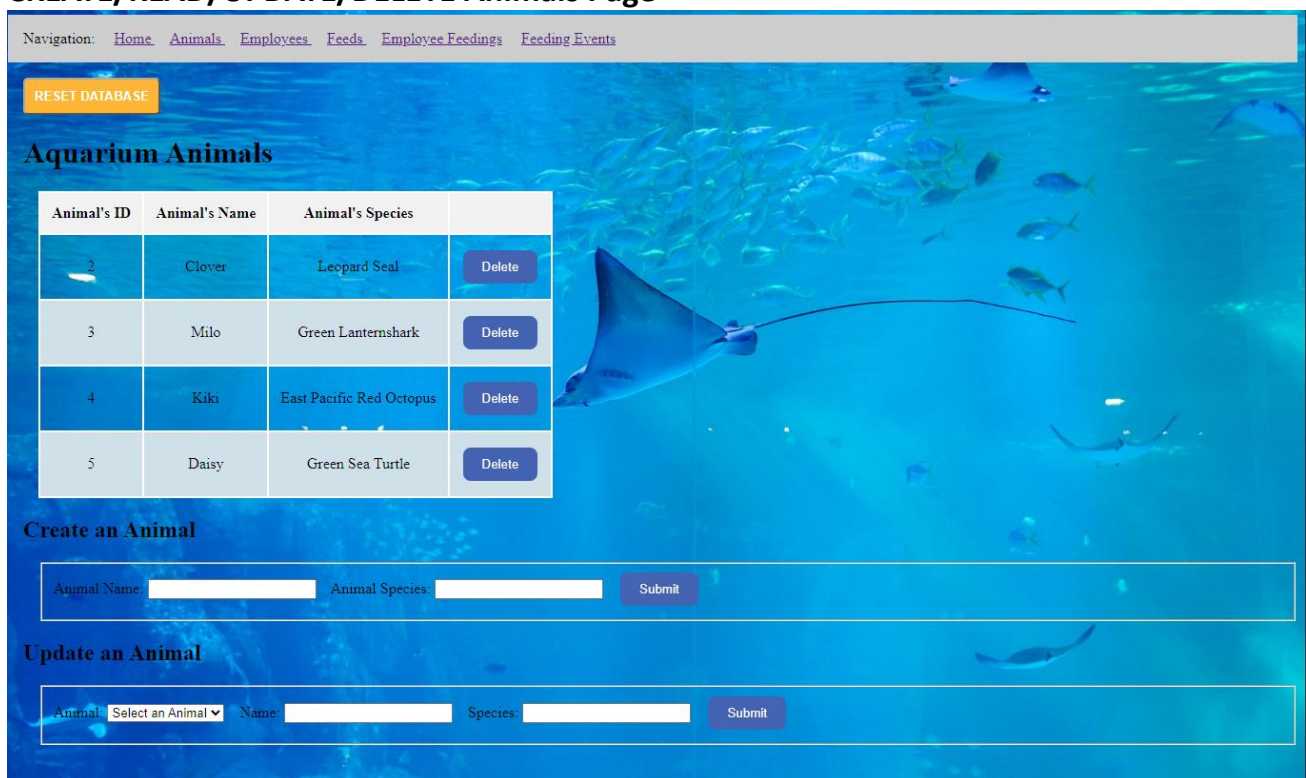
SCREENSHOTS OF USER INTERFACE

Home Page

Reset Database



CREATE/READ/UPDATE/DELETE Animals Page



CREATE/READ/UPDATE/DELETE Employees Page

Navigation: [Home](#) [Animals](#) [Employees](#) [Feeds](#) [Employee Feedings](#) [Feeding Events](#)

[RESET DATABASE](#)

Aquarium Employees

Employee ID	First Name	Last Name	Age	Employment Type	
1	Steve	Irwin	45	Full-time	Delete
2	Robert	Jackson	22	Part-time	Delete
3	Elizabeth	Raynes	35	Full-time	Delete
4	Mary	Jones	20	Part-time	Delete

Create an Employee

Employee First Name: Employee Last Name: Employment Status: Age: [Submit](#)

Update an Employee

Employee: First Name: Last Name: Employment Status: Age: [Submit](#)

CREATE/READ/UPDATE/DELETE Feeds Page

Navigation: [Home](#) [Animals](#) [Employees](#) [Feeds](#) [Employee Feedings](#) [Feeding Events](#)

[RESET DATABASE](#)

Aquarium Feeds

Feed ID	Feed Name	Feed Description	
1	Brine Shrimp Blend	A protein rich feed for carnivore animals.	Delete
2	Frozen Fish and Supplement Mix	A protein, vitamin and mineral rich feed for omnivore animals.	Delete
3	Plant Flake	A fiber rich feed for herbivore animals.	Delete

Create a Feed

Feed Name: Feed Description: [Submit](#)

Update a Feed

Feed Name: Name: Description: [Submit](#)

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CREATE/READ/UPDATE/DELETE EmployeeFeedings Page

Update M:N relationship

Delete from M:N

Navigation: [Home](#) [Animals](#) [Employees](#) [Feeds](#) [Employee Feedings](#) [Feeding Events](#)

[RESET DATABASE](#)

Employee Feedings

Employee Feeding ID	Employee ID	Employee Full Name (Helper Column)	Feeding Event ID	Feeding Event Time (Helper Column)	
3	1	Steve Irwin	2	2026-07-21 10:15:00	Delete
4	3	Elizabeth Raynes	3	2026-07-21 10:40:00	Delete
5	4	Mary Jones	3	2026-07-21 10:40:00	Delete
6	4	Mary Jones	4	2026-07-21 11:00:00	Delete
7	2	Robert Jackson	5	2026-07-21 11:00:00	Delete

Create a Employee Feeding Event

Employee ID: [Select an Employee ID](#) Feeding Event ID: [Select a feeding event ID](#) [Submit](#)

Update a Employee Feeding Event

[Select Employee Feeding](#) [Select an Employee Feeding ID](#) [Select Employee](#) [Select an Employee ID](#) [Select Feeding Event](#) [Select a Feeding Event ID](#) [Submit](#)

CREATE/READ/UPDATE/DELETE FeedingEvents Page

Navigation: [Home](#) [Animals](#) [Employees](#) [Feeds](#) [Employee Feedings](#) [Feeding Events](#)

[RESET DATABASE](#)

Feeding Events

Feeding Event ID	Feed Time	Animal ID	Animal Name (Helper Column)	Feed ID	Feed Name (Helper Column)	
2	2026-07-21 10:15:00	2	Clover	1	Brine Shrimp Blend	Delete
3	2026-07-21 10:40:00	3	Milo	1	Brine Shrimp Blend	Delete
4	2026-07-21 11:00:00	4	Kiki	2	Frozen Fish and Supplement Mix	Delete
5	2026-07-21 11:00:00	5	Daisy	3	Plant Flake	Delete

Create a Feeding Event

Feeding Event Time: Select an Animal ID: [Select an Animal's Name](#) Select a Feed ID: [Select a Feed](#) [Submit](#)

Update a Feeding Event

[Select Feeding Event](#) [Select a Feeding event's ID](#) Update Feeding Event Time: Animal: [Select an Animal's ID](#) Feed: [Select a Feed ID](#) [Submit](#)

Citations

- [1] <https://nyaquarium.com/animals/our-animals>
- [2] <https://www.mysticaquarium.org/animals-and-exhibits/our-animals/>

Academic Integrity Statement

Kelli Muldoon and Reza Can Cihangir pledge that no AI was used during the preparation of this final report.